

EXP NO: 9

Basic Calculator App Using Android UI Controls

AIM:

Develop an Android application using controls like Button, TextView, EditText for designing a calculator having basic functionality like Addition, Subtraction, multiplication, and Division.

CODE:

MainActivity.kt

```
package com.example.myapplication_karthick_79

import android.os.Bundle
import android.view.View
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    private lateinit var display: EditText
    private var currentInput: String = ""
    private var operand1: Double = 0.0
    private var operand2: Double = 0.0
    private var operator: String = ""

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        display = findViewById(R.id.display)

        // Number buttons
        setButtonClickListener(R.id.button0)
        setButtonClickListener(R.id.button1)
        setButtonClickListener(R.id.button2)
        setButtonClickListener(R.id.button3)
        setButtonClickListener(R.id.button4)
        setButtonClickListener(R.id.button5)
        setButtonClickListener(R.id.button6)
        setButtonClickListener(R.id.button7)
        setButtonClickListener(R.id.button8)
        setButtonClickListener(R.id.button9)

        // Operator buttons
        setOperatorClickListener(R.id.buttonAdd, "+")
        setOperatorClickListener(R.id.buttonSubtract, "-")
        setOperatorClickListener(R.id.buttonMultiply, "*")
        setOperatorClickListener(R.id.buttonDivide, "/")
    }
}
```

```

// Clear button
findViewById<Button>(R.id.buttonClear).setOnClickListener {
    currentInput = ""
    operand1 = 0.0
    operand2 = 0.0
    operator = ""
    display.setText("")
}

// Equal button
findViewById<Button>(R.id.buttonEqual).setOnClickListener {
    if (operator.isEmpty()) return@setOnClickListener

    try {
        operand2 = currentInput.toDouble()

        val result = when (operator) {
            "+" -> operand1 + operand2
            "-" -> operand1 - operand2
            "*" -> operand1 * operand2
            "/" -> {
                if (operand2 == 0.0) {
                    Toast.makeText(this@MainActivity, "Cannot divide by zero",
Toast.LENGTH_SHORT).show()
                    return@setOnClickListener
                }
                operand1 / operand2
            }
            else -> 0.0
        }

        display.setText(result.toString())
        operand1 = result // Update operand1 for subsequent calculations
        operator = ""
        currentInput = result.toString()
    } catch (e: Exception) {
        Toast.makeText(this@MainActivity, "Error: Invalid Input",
Toast.LENGTH_SHORT).show()
    }
}

// Set up number button listeners
private fun setButtonClickListener(buttonId: Int) {
    val button: Button = findViewById(buttonId)
    button.setOnClickListener {
        currentInput += button.text.toString()
        display.setText(currentInput)
    }
}

// Set up operator button listeners
private fun setOperatorClickListener(buttonId: Int, op: String) {
    val button: Button = findViewById(buttonId)

```

```

        button.setOnClickListener {
            if (currentInput.isNotEmpty()) {
                operand1 = currentInput.toDouble()
                currentInput = ""
                operator = op
            }
        }
    }
}

// Optional: Add functionality to handle decimal point (if needed)
private fun isDecimalPointValid(): Boolean {
    return !currentInput.contains(".")
}
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/calculatorLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/display"
        android:layout_width="match_parent"
        android:layout_height="80dp"
        android:background="#000000"
        android:textColor="#FFFFFF"
        android:textSize="28sp"
        android:gravity="end|center_vertical"
        android:inputType="none"
        android:focusable="false"
        android:clickable="false"
        android:padding="12dp"
        android:layout_marginBottom="12dp" />

    <!-- Row 1: 7 8 9 / -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:id="@+id/button7" style="@style/CalcButton" android:text="7" />
        <Button android:id="@+id/button8" style="@style/CalcButton" android:text="8" />
        <Button android:id="@+id/button9" style="@style/CalcButton" android:text="9" />
        <Button android:id="@+id/buttonDivide" style="@style/CalcButton" android:text="/" />
    </LinearLayout>

    <!-- Row 2: 4 5 6 * -->
    <LinearLayout

```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:id="@+id/button4" style="@style/CalcButton" android:text="4" />
        <Button android:id="@+id/button5" style="@style/CalcButton" android:text="5" />
        <Button android:id="@+id/button6" style="@style/CalcButton" android:text="6" />
        <Button android:id="@+id/buttonMultiply" style="@style/CalcButton" android:text="*" />
    </LinearLayout>

```

<!-- Row 3: 1 2 3 - -->

```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
    <Button android:id="@+id/button1" style="@style/CalcButton" android:text="1" />
    <Button android:id="@+id/button2" style="@style/CalcButton" android:text="2" />
    <Button android:id="@+id/button3" style="@style/CalcButton" android:text="3" />
    <Button android:id="@+id/buttonSubtract" style="@style/CalcButton" android:text="-" />
</LinearLayout>

```

<!-- Row 4: 0 C = + -->

```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
    <Button android:id="@+id/button0" style="@style/CalcButton" android:text="0" />
    <Button android:id="@+id/buttonClear" style="@style/CalcButton" android:text="C" />
    <Button android:id="@+id/buttonEqual" style="@style/CalcButton" android:text="=" />
    <Button android:id="@+id/buttonAdd" style="@style/CalcButton" android:text="+" />
</LinearLayout>

```

</LinearLayout>

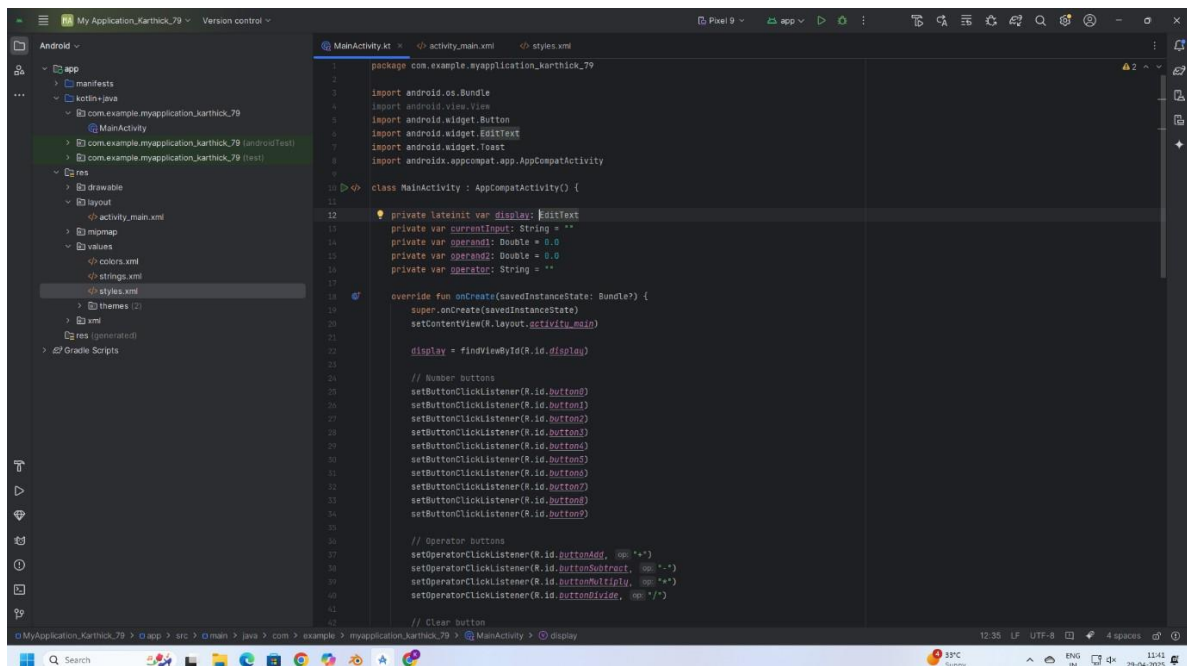
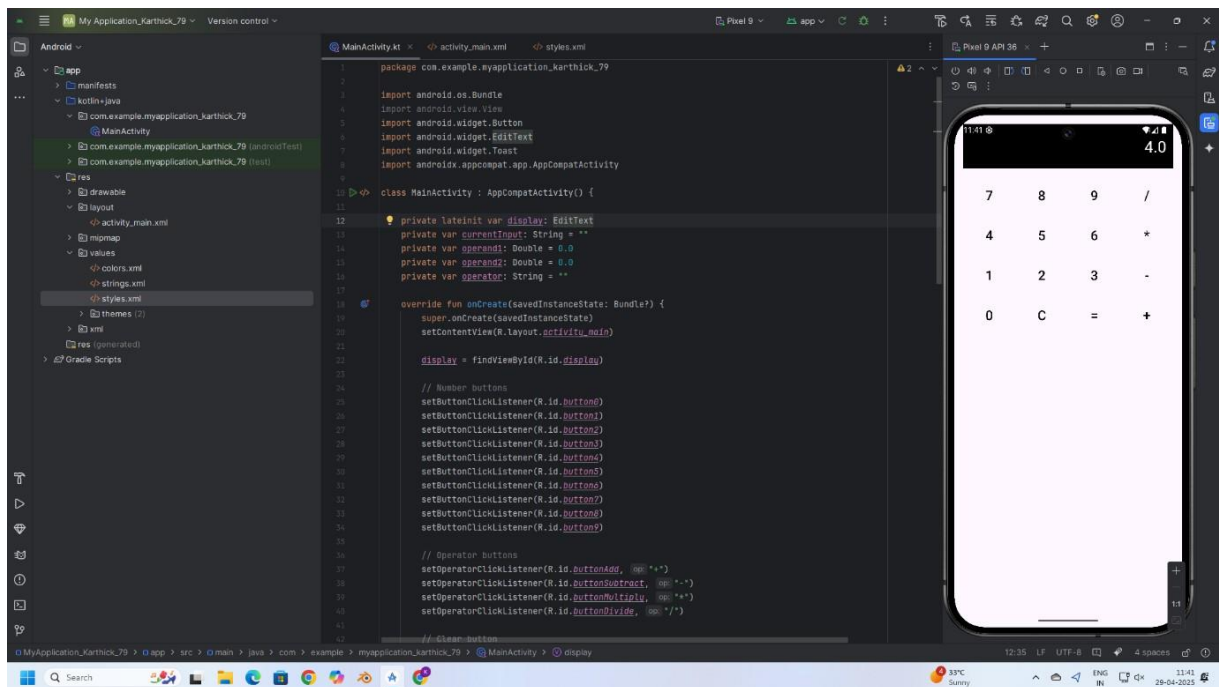
styles.xml

```

<resources>
<style name="CalcButton">
<item name="android:layout_width">0dp</item>
<item name="android:layout_height">wrap_content</item>
<item name="android:layout_weight">1</item>
<item name="android:padding">16dp</item>
<item name="android:textSize">24sp</item>
<item name="android:background">?attr/selectableItemBackground</item>
<item name="android:textColor">#000000</item>
</style>
</resources>

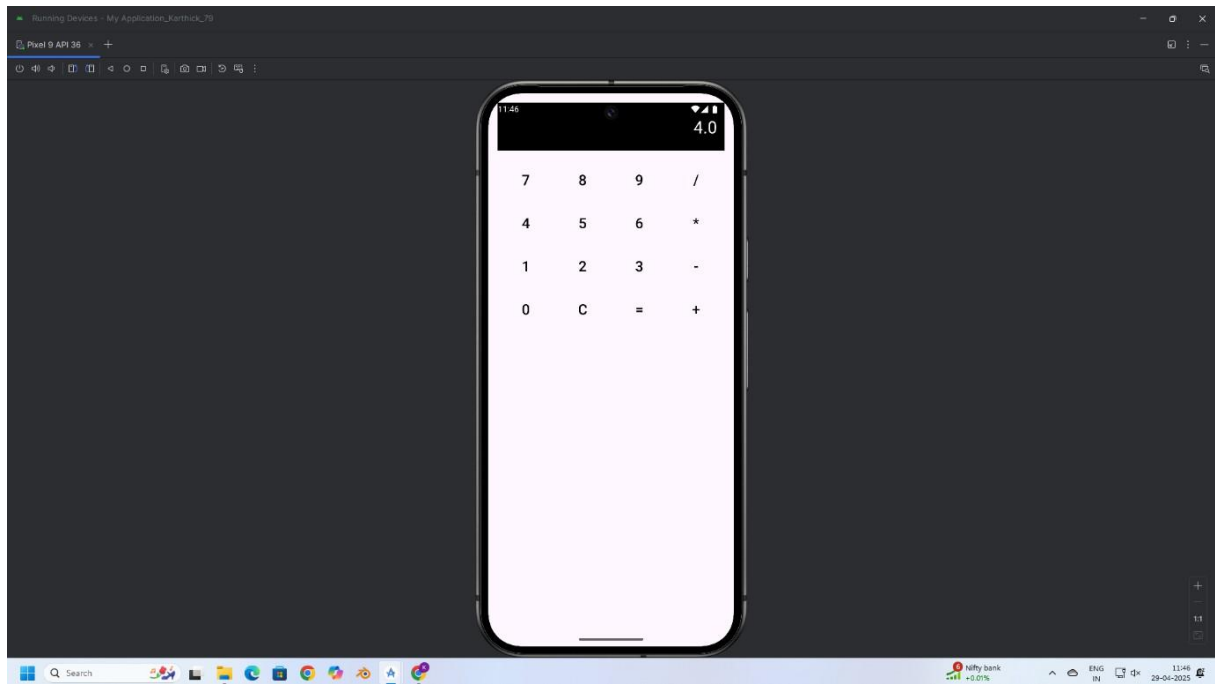
```

OUTPUT:



```
1 package com.example.myapplication_karthick_79
2
3 import androidx.appcompat.app.AppCompatActivity
4 import android.os.Bundle
5 import android.widget.Button
6 import android.widget.TextView
7
8 class MainActivity : AppCompatActivity() {
9     private var result: Double = 0.0
10
11     override fun onCreate(savedInstanceState: Bundle?) {
12         super.onCreate(savedInstanceState)
13         setContentView(R.layout.activity_main)
14
15         // Initialize TextView
16         textView = findViewById<TextView>(R.id.textView)
17
18         // Initialize Buttons
19         button1 = findViewById<Button>(R.id.button1)
20         button2 = findViewById<Button>(R.id.button2)
21         button3 = findViewById<Button>(R.id.button3)
22         button4 = findViewById<Button>(R.id.button4)
23         button5 = findViewById<Button>(R.id.button5)
24         button6 = findViewById<Button>(R.id.button6)
25         button7 = findViewById<Button>(R.id.button7)
26         button8 = findViewById<Button>(R.id.button8)
27         button9 = findViewById<Button>(R.id.button9)
28         button10 = findViewById<Button>(R.id.button10)
29
30         // Set ClickListeners
31         button1.setOnClickListener { calculate(1) }
32         button2.setOnClickListener { calculate(2) }
33         button3.setOnClickListener { calculate(3) }
34         button4.setOnClickListener { calculate(4) }
35         button5.setOnClickListener { calculate(5) }
36         button6.setOnClickListener { calculate(6) }
37         button7.setOnClickListener { calculate(7) }
38         button8.setOnClickListener { calculate(8) }
39         button9.setOnClickListener { calculate(9) }
40         button10.setOnClickListener { calculate(10) }
41     }
42
43     private fun calculate(operation: Int) {
44         when (operation) {
45             1, 2, 3, 4, 5, 6, 7, 8, 9, 10 -> {
46                 // Perform calculation based on button pressed
47                 // For example, if button 1 is pressed, result becomes 1.0
48                 result = 0.0
49                 result = result * operation
50                 textView.text = result.toString()
51             }
52         }
53     }
54 }
```

```
1 <resources>
2     <style name="CalcButton">
3         <item name="android:layout_width">0dp</item>
4         <item name="android:layout_height">wrap_content</item>
5         <item name="android:padding">10dp</item>
6         <item name="android:textSize">20sp</item>
7         <item name="android:background">?attr/selectableItemBackground</item>
8         <item name="android:textColor">#000000</item>
9     </style>
10 </resources>
```

**RESULT:**

Thus, a basic calculator application was successfully developed using Android controls like Button, TextView, and EditText to perform addition, subtraction, multiplication, and division operations.